

Putting Myths to Rest: Spectral Analysis of PGA Tour Metrics

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1 Introduction

Drive for show; putt for dough. In the world of professional golf, the ability to master the greens is not just an advantage but often the linchpin of championship success. Putting demands precision, cold-bloodedness, and a little bit of luck. As such it commands a pivotal role in the narrative of the sport, with the putter often deemed the ultimate equaliser or differentiator between contenders. Successful putts can catalyse victories whilst missed opportunities can doom even the most skilled tee-to-green players to defeat by undermining all of their self-trust. They say that to master the sport, you must work your way out from hole to tee. If you manage to hole putts easily, the rest of your game can loosen up as your confidence sees itself unleashed. After all, a 3-footer holds the same weight on the scorecard as a 400 yard drive. It follows that importance of putting has led to the popular belief that professional golfers can enter periods of “hot” streaks: time periods where the cup gets larger and putts seem infallibly guided to the bottom of the hole. Successive wins and remarkable career highlights are often attributed to said phenomenon.

The notion of *streakiness* – successive periods of above-average success punctuated by either the cumulative effect of confidence or by sheer providence – is a captivating element across sports disciplines. It feeds into the importance of the mental game in competitive sports where belief and positive imagery can be the difference between triumph or loss. In the case of golf, if you see yourself missing putts, you’re less likely to trust the process, release and let your athletic ability do the work. On the other hand, if you can picture the success of your decisions and effort, you are more likely to visualise and commit to your shots, letting muscle-memory lead you to success. This phenomenon feeds into the drama of athletic competition, shaping how spectators and commentators perceive player performance.

In sports like basketball, the “*hot hand*” phenomenon has been researched and debated. A study which analysed the NBA Long Distance Shootout suggested that this effect is more a narrative construct than a predictable pattern, challenging the statistical reality of such streaks in professional basketball (Koehler, J. J., & Conley, C. A., 2003).

There has been limited statistical research done on this phenomenon. Prior literature seems to suggest that in the late 90s, a small number of PGA Tour participants experienced statistically significant streaky play (Yousefi, K. & Swartz, T., 2013), however, traditionally golf analytics have skirted the fringes of this phenomenon, failing to definitively squash or positively confirm that streakiness indeed exists generally in the sport.

In this practice of inference we will use spectral domain analysis to break down more recent PGA Tour time series data of putting performance to examine patterns that might elude simple statistical models with which to forecast putting performance, and therefore success for different players.

The paper will explore PGA Tour putting metrics to rigorously assess the hypothesis of streakiness in golf putting. We aim to use spectral domain analysis to ascertain whether the streaks we “observe” or believe exist, are grounded in real phenomena underpinned by statistical regularities or whether they are merely the byproducts of humans seeking pattern and structure where it doesn’t exist. The implications of this paper extend beyond academic interest as they challenge entrenched narratives in professional golf and may help guide enthusiasts, in their betting efforts when their favourite players seem to get “*hot*”. The findings of this research could potentially help deconstruct the notion of streakiness in other sports as we slowly, sport by sport, statistically debunk these myths. We seek to change how performance in professional sports is comprehended paving the way for a new era of sports analytics grounded in scientific rigour. In section 2, we will proceed by detailing our data set, and how it was attained. In section 3, we will perform the spectral domain analysis of our data and in our final section, 4, we will present the implications of our results and findings.

2 Methods

Our dataset comprised of approximately 37,000 event results belonging to approximately 500 golfers on the PGA Tour from 2015 to 2022. The data was originally taken from the

PGA Tour’s website which records and uploads relevant statistics for all the players participating in any of their events.

For this study, we chose to focus on Strokes Gained Putting average per tournament (usually Stroke Gained averages computed over 4 tournament rounds) of highly performing PGA Tour players (Yousefi, K. & Swartz, T., 2013). This is a statistical metric used in golf to evaluate a player’s putting performance against a professional standard.

Strokes gained calculates the average number of strokes a pro golfer would take to hole out from a specific distance, and compares this with the actual strokes taken by a given player. It essentially measures the quality of a putt relative to the field, considering both the distance of the putt and the expected number of strokes a professional would make from that distance. A positive value indicates better performance than the average professional, whilst a negative value suggests the opposite. Generally, a Strokes Gained Putting value between 0.5 and 1.0 is considered excellent as it suggests that the golfer is consistently performing better than the average professional standards on greens. A SG Putting value of 1.0 means that the golfer is gaining one stroke on the field average per round due to their putting, and indicates a significant advantage in putting performance compared to the average professional golfer. Since we are evaluating the highest performing level of Tour players, we expect to see positive values on average.

We chose to narrow down the data set into a few randomly selected highly ranked individuals with over 100 events recorded who have been in the top 30 of the Official World Golf Ranking in the most recent years. We ended up choosing the following players: Adam Scott, Billy Horschel, Brandt Snedeker, Brian Harman, Bubba Watson, Danny Lee, Dustin Johnson, Francesco Molinari, Gary Woodland, Sergio Garcia, Xander Schauffele and Jon Rahm.

Focusing statistical analyses on individuals in the top 30 of the OWGR can yield several advantages. Players within this performance category record more PGA tournaments a year (which makes the data set more robust and complete) and are notably perceived to exhibit streaky behaviour. Enthusiasts thinking about the PGA Tour tend to be more interested in the behaviour of the most successful. Analyzing this elite group

allows for comparative insights into the best practices and strategies in the sport, offering a clear view of what distinguishes top performers from the rest. This approach can uncover trends and factors that contribute significantly to high-level success. We will be using R, to facilitate this statistical analysis.

3 Data Analysis

Before analysing our data, we ensured our time series met our modelling assumptions, that is, we ensured that our data was weakly stationary. The assumption of stationarity is met when $\mu_t = \mu$ and $\gamma(t, t + h) = \gamma(h)$, so when the mean of our process and the covariance structure is independent of time. This ensures that the properties of our time series are not dependent on the period of time in which we observe it. The covariance we calculate is dependent on the distance/lag between observations instead. This assumption is essential as modelling relies on the assumption that patterns observed in the past will continue to be observed in the future.

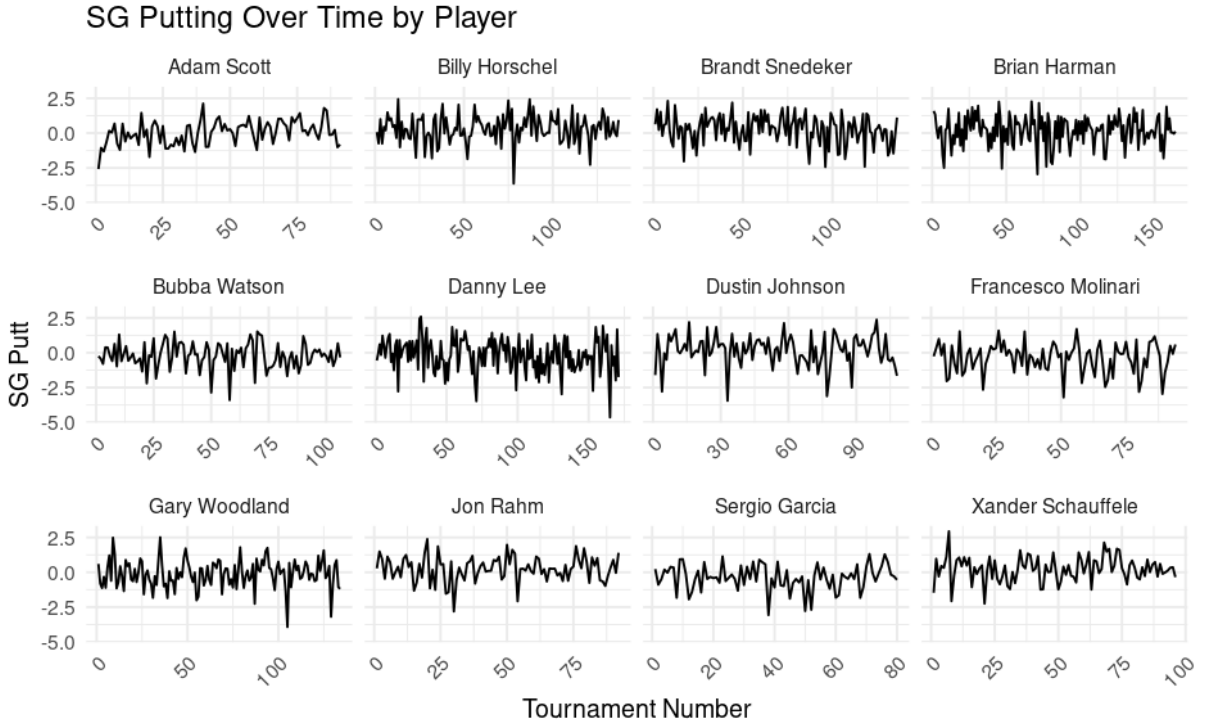


Figure 1: Raw Time Series

Upon observation, the graphs of our raw time series, indicate that the data for all

our players is indeed centred around the mean and it seems to be homoscedastic, meeting our basic assumptions. So we can proceed with our analysis.

We proceed to compute the ACF, autocorrelation function, which measures and plots the average correlation between data points in our time series and previous values for different lag quantities. If indeed streakiness is observed (whether for good tournament streaks or bad), we would hope to see high ACF at lags other than 0. This would imply, that if I play well this event, I can hope to play well next event and in the near events to come.

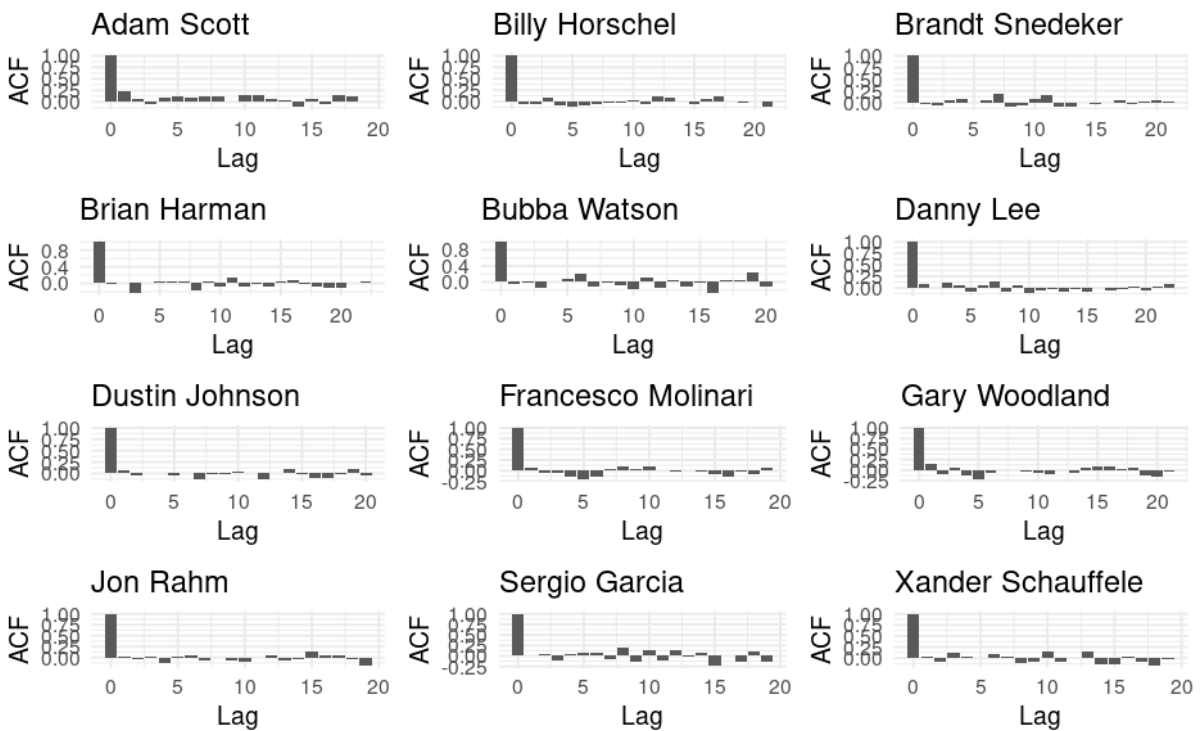


Figure 2: ACF of PGA Tour Players SG Putting

In Figure 2, however, we observe that pretty much for every player, all signal drops after lag 0. That is, upon first inspection, there seems to be insignificant correlation between the putting performance at different events. Good and bad putting doesn't seem to carry over; any semblance seems to be explained by white noise, that is, error! We turn to the scaled periodogram to see if we can observe anything different. From the ACF, we can't reject the hypothesis of white noise in streakiness. Streakiness is most important in weeks next to each other, and we have already seen significant statistical significance to debunk this, however maybe there is seasonality or some other sort of signal that we

can use to predict a player's putting performance.

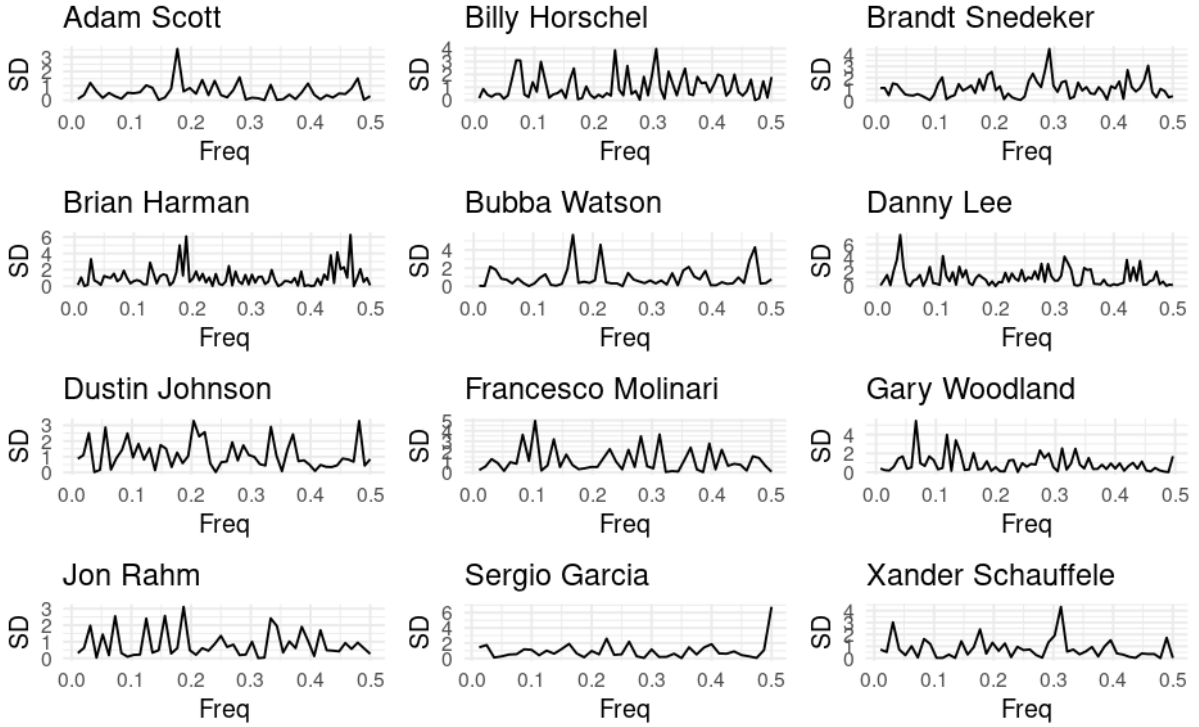


Figure 3: Scaled Periodogram of PGA Tour Players SG Putting

The scaled periodogram helps us identify and quantify the frequency components within a time series, showing how the data's variance is distributed across different frequencies. We plot all frequencies up to $1/2$, the Nyquist frequency, which is the minimum rate at which a signal can be sampled without introducing aliasing errors. It helps us uncover cyclical behaviours from white noise, and distinguishing signal from noise. In our case, we fail to see a significant pattern in the periodograms which could very much again indicate that variation from putting performance is nothing but white noise. We do observe Sergio Garcia's anomalous distribution with a peak at frequency $1/2$, and quiet signal for the remainder of frequencies. His good putting tournaments are followed by bad ones and viceversa. To see this result is unsurprising as at a 95% confidence level, we expect to see $1/20$ anomalous results. Our other observations pretty consistently seem to indicate that there exists insignificance evidence to suggest that there's a periodic nature or streaky nature in a player's putting performance.

Finally, to convince ourselves that there really seems to be no statistical evidence to suggest that there's structure in strokes gained putting over different events, we look

at the cumulative periodogram of our time series which helps us check whether this is indeed white noise. These plots show the cumulative sum of the normalised periodogram against all the frequencies of interest. For a purely white noise process, the cumulative periodogram should resemble a straight $y = x$ line, indicating the uniform distribution of power across different frequencies. The plots can facilitate more confidence in the fact that the ACF seems to be 0 at all values of lag h other than 0 itself, indicating no correlation between different events in time.

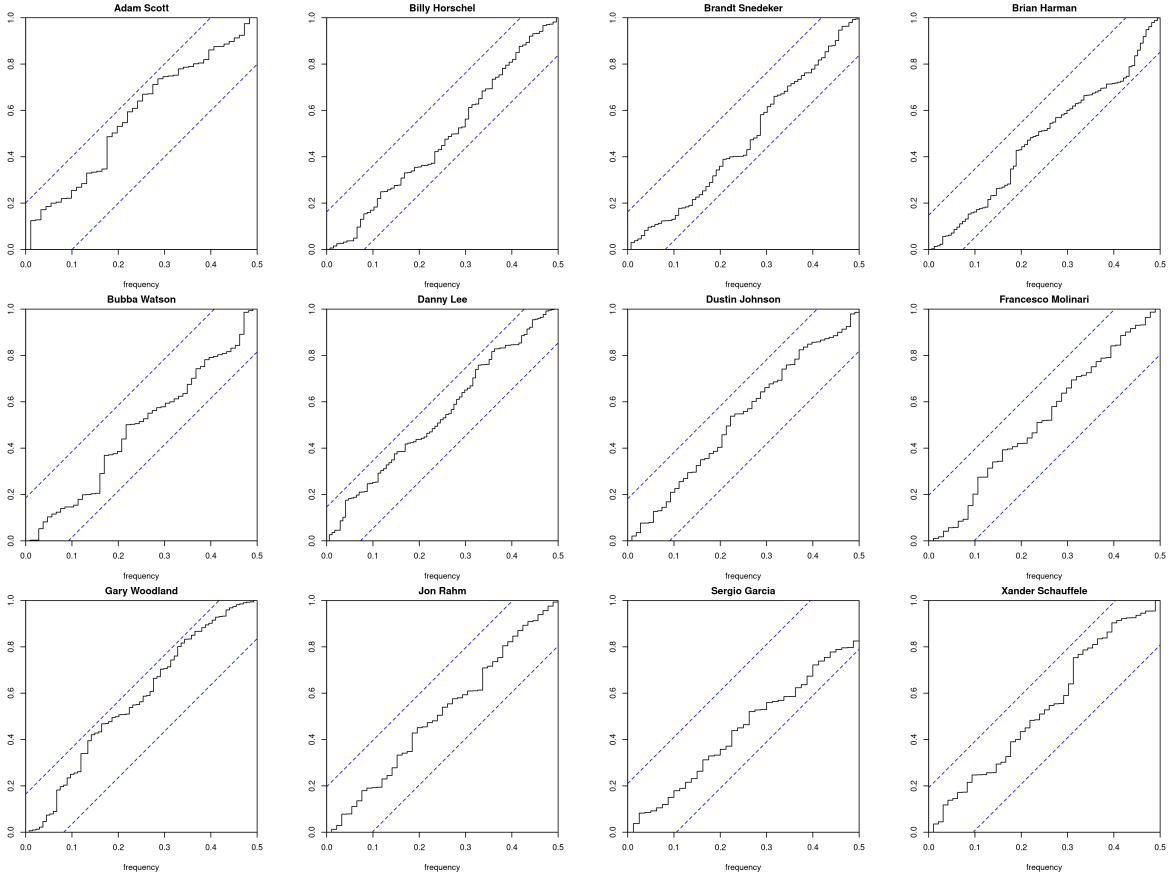


Figure 4: Cumulative Periodogram of PGA Tour Players SG Putting

We can observe that for all our players, our cumulative periodograms seem to resemble $y = x$ lines within 95% confidence intervals, implying that indeed, there is insufficient evidence to suggest that putting performance is not white noise and to imply that the “*streakiness*” effect is real.

4 Conclusion

In this paper, we have used spectral domain analysis tools to examine the supposed existence of “streakiness” in the putting performance of elite PGA Tour golfers. Through detailed statistical analysis, including the evaluation of autocorrelation functions and periodograms, we have seen that the data seems to suggest a predominant presence of white noise in putting performance, suggesting that the variations in putting success amongst the studied golfers does not exhibit significant serial correlation or cyclical patterns that could be systematically exploited for predictive advantage.

We have dispelled the popular notion amongst players and fans that golfers can experience predictable “*hot*” streaks. This very much aligns with some literature on the topic in other sports such as basketball. We should reassess strategies dependent on the perceived momentum within tournaments and rest assured that if we play golf badly today, we won’t necessarily in our next tournament. We should approach each golf tournament as a new event, independent of previous outcomes.

It is important before concluding the paper that we address some weaknesses inherent in this study. We have taken different events as our distinct time points. However, the data we have utilised goes from 2015 to 2022, and the spacing between events and seasons is not consistent. Players may take time off from events, and surely off-season could potentially cool off or warm up players as they take months off at a time to prepare and improve for the next season.

Future research might expand on this study by analysing over periods of time where players are in season and expected to compete every week, this would more closely meet time series assumptions. Similarly it would be interesting to evaluate a larger range of players of differing levels of skill and performance. Similar methodologies could be applied investigate “streakiness” in other aspects of the sport such as driving or recovery play, or even putting but within a round itself.

Ultimately, what we have argued here is that you don’t have to worry. Just because you played badly last week, doesn’t necessarily mean you will this weekend. All hope is not lost.

References

- [1] Koehler, J. J., & Conley, C. A. (2003). The “Hot Hand” Myth in Professional Basketball. *Journal of Sport and Exercise Psychology*, 25(2), 253-259. Retrieved May 10, 2024, from <https://doi.org/10.1123/jsep.25.2.253>
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